Project :)

Galaxy Classification ~~project~~

17 August 2025

Part I
Module 1: Task 1

1 Dark Matter Framework

1.1 The role of dark matter halos into the cosmic web structure we observe today

Every galaxy, including our own Milky Way, is thought to reside at the center of a much larger, invisible dark matter halo. Here's their crucial role:

- **Gravitational Anchors:** Dark matter halos provide the dominant gravitational pull necessary to hold galaxies together. The visible matter in galaxies (stars, gas, dust) accounts for only a fraction of the total mass needed to explain their observed rotation and dynamics. Dark matter, which doesn't interact with light, provides this missing mass, extending far beyond the visible galactic disk.

- **Galaxy Formation Sites:** In the early universe, tiny density fluctuations in the dark matter grew due to gravity, collapsing to form these halos. These overdense regions then attracted ordinary baryonic matter (gas and dust), which subsequently cooled and condensed within the halo's gravitational well to form stars and, eventually, galaxies.

- **Influence on Galaxy Evolution:** The growth and evolution of galaxies are intimately tied to the growth of their dark matter halos through hierarchical merging. As smaller halos merge to form larger ones, they bring along their associated galaxies, leading to galaxy mergers and the assembly of larger galactic structures. The properties of a galaxy (like its size, stellar mass, and star formation rate) are strongly correlated with the properties of its host dark matter halo.

1.2 How primordial density fluctuations evolved into the cosmic web structure we observe today

The universe we see today grew from tiny, almost imperceptible "seeds" in the very early universe, known as primordial density fluctuations. Imagine the early universe as an incredibly smooth, hot soup of particles, but with very, very slight variations in how dense it was. Some regions were a tiny bit denser, others a tiny bit emptier. These minute differences are the "primordial density fluctuations."

The key to understanding how these tiny differences grew into the cosmic web lies in anisotropic gravitational collapse. "Gravitational collapse" simply means that matter, due to its own gravity, pulls itself together. "Anisotropic" means this pulling didn't happen uniformly in all directions. Instead, it was stronger in some directions than others.

The exact shape a region would take– whether it became a vast empty space (a void), a thin flat structure (a sheet or "pancake"), a long string-like structure (a filament), or a dense clump (a cluster) – depended on how gravity was pulling on it from different directions. This "pull" can be mathematically described by something called a deformation tensor. This is like a map that tells us how much and in what ways a particular region of space is being stretched or squeezed by gravity. The "fate" of a region is determined by the eigenvalues of this deformation tensor.

- If all the "squeezing" forces were weak or pushing outwards, the region became a void – a large, underdense empty space.
- If gravity was strong enough to squeeze in one direction, it formed a sheet or "pancake."
- If it squeezed in two directions, it formed a filament – a cosmic thread.
- And if gravity squeezed strongly from all three directions, it resulted in a cluster – the densest knots in the cosmic web.

This process followed a specific sequence over billions of years. Regions first flattened into sheets, then these sheets broke down and condensed further into filaments, and finally, where multiple filaments intersected, the densest clusters of galaxies formed.

So, the faint ripples of density in the early universe were the blueprints. Over cosmic time, gravity, acting in a non-uniform way, pulled matter together along these blueprints, creating the magnificent, interconnected cosmic web of galaxies, filaments, sheets, and voids that we observe across the universe today.

2 Baryonic Physics

Baryonic physics helps us understand how ordinary matter behaves, especially how galaxies form and evolve. This involves processes like gas cooling and its condensation within dark matter halos, eventually leading to star formation.

2.1 Gas cooling processes

Gas in the outer space maybe extremely hot and sometimes loses energy by emitting light which is known as radiative cooling. The way it cools depends on its temperature:

- High Temperatures (above 107 K): In extremely hot, ionized gas, like in galaxy clusters, free electrons scatter off atomic nuclei and emit X-rays. This is known as Bremsstrahlung or free-free emission.

- Intermediate temperatures (104–107 K): As gas cools, electrons can collide with atoms or ions, exciting them. When these excited particles return to a lower energy state, they release energy as specific wavelengths of light (spectral lines). This is called collisional excitation and radiative de-excitation (or line cooling) and is very efficient, especially when heavier elements are present.

- Low Temperatures (below 104 K): At these temperatures, atoms can combine to form molecules (like hydrogen and carbon monoxide). These molecules can cool the gas further by emitting light when their rotational and vibrational energy levels are excited by collisions.

2.2 Condensation within Dark Matter Halos

Dark matter halos are invisible structures that provide the gravitational structures for galaxies -

- Infall and Heating: As gas falls into a dark matter halo, it gets compressed and heated to a high temperature due to gravitational shocks.

- Cooling and Collapse: The hot gas then starts to cool through the processes described above. Denser regions cool faster, lose pressure, and are pulled further inward by gravity. This leads to a continuous flow of cooling gas toward the halo's center.

- Condensation: This cooling and inward flow causes the gas to condense, becoming denser and colder. The more concentrated the baryonic matter becomes, the more it pulls dark matter inward, increasing its density in the central regions of halos.

2.3 Mechanisms Triggering Star Formation in Early Galaxies

Star formation is a direct consequence of gas condensation. When gas cools to sufficiently low temperatures (below 104 K) and reaches high densities, it becomes gravitationally unstable. This instability causes the gas to fragment into even denser clumps, which then collapse to form stars.

Cosmological simulations show that this condensation of gas leads to a steeper dark matter profile in the inner regions of halos compared to simulations without gas cooling. Furthermore, this process makes dark matter halos significantly more spherical than if gas cooling were not considered. The continued cooling and star formation, even at lower redshifts (later times in the universe's history), contributes to the growth of both baryonic matter and dark matter density in the central regions of these systems.

3 Stellar Evolution

At the dawn of the Universe, approximately 380,000 years after the Big Bang, the cosmos was a much simpler place. Its elemental composition was nearly primordial, consisting almost entirely of hydrogen and helium, along with trace amounts of lithium. Crucially, heavier elements, which astronomers collectively refer to as "metals," were absent.

The formation of the very first stars, Population III stars, occurred within this metal-free environment. Their birthing conditions were dictated by the limited cooling mechanisms available: primarily molecular hydrogen (H2) and hydrogen deuteride (HD). These molecules are less efficient coolants than metals, meaning that the gas clouds needed to be substantially more massive to overcome thermal pressure and gravitationally collapse. Consequently, Pop III stars are theorized to have been exceptionally massive (ranging from tens to hundreds of solar masses, with some models suggesting up to 500 solar masses), leading to a "top-heavy" initial mass function (IMF), where very massive stars were abundant. A direct consequence of their immense mass was their extremely short lifespans, typically on the order of a few million years (about 10^6 years).

The death of these trailblazing Pop III stars was the catalyst for the Universe's chemical enrichment. Many of the most massive Pop III stars would have ended their lives as pair-instability supernovae (PISN), particularly those in the 140-260 solar mass range. These cataclysmic explosions ejected newly synthesized metals—such as oxygen, carbon, silicon, and iron—into the surrounding interstellar and intergalactic medium (ISM and IGM).

Simulations indicate that this metal enrichment process is remarkably rapid within high-density regions of the early Universe. While the distribution of metals is initially very "patchy" (uneven), with pristine regions coexisting with enriched ones, the local metallicity around nascent star-forming sites is quickly boosted above the threshold required for Pop II star formation. Consequently, the Pop III star formation regime lasts for a very short period in cosmic history, with its average contribution to the total star formation rate density rapidly dropping off (below 10^{-3}–10^{-2}) shortly after the first supernovae begin to explode.

In summary, Pop III stars, though short-lived, were the cosmic pioneers. Their explosive deaths provided the essential "metals" that transformed the Universe's gas, enabling more efficient cooling and fragmentation. This, in turn, allowed for the birth of Pop II stars with a more familiar mass distribution, fundamentally altering the trajectory of galactic chemical evolution and setting the stage for the complex, chemically diverse galaxies we observe today.

4 Feedback Processes

4.1 Supernovae explosions

Supernovae influence Star Formation process mainly by:

- Heating and Dispersing Gas: When a massive star explodes, it releases an enormous amount of energy. This energy heats up the nearby cool gas and dust in the interstellar medium (ISM) – the material between stars in a galaxy. This hot gas expands and spreads out, making it harder for new clumps of gas to form stars. This process is called "supernova feedback".
- Balancing Energy: The rate at which stars form is a balance: On one side, energy is lost when cold gas clouds bump into each other and dissipate energy. On the other side, supernovae constantly inject energy into the ISM. This continuous energy input from supernovae helps regulate how quickly new stars can form.
- Conversely, the shockwaves generated by supernova explosions can compress nearby gas clouds. This compression can sometimes trigger the collapse of these clouds, leading to the formation of new stars.

Supernovae don't just affect immediate star formation; they also play a crucial role in the long-term development and appearance of galaxies:

- Gas Expulsion and Galactic Winds: The hot gas created by supernovae can become so energetic that it escapes the galaxy entirely, forming what's called a galactic wind. This is particularly effective in smaller galaxies. By blowing gas out, supernovae remove the "fuel" needed for new stars, thus limiting the total number of stars a galaxy can form over its lifetime. For a Milky Way-like galaxy, this model predicts around 20-30
- Galactic Fountains: If the hot gas doesn't escape the galaxy entirely (meaning its temperature isn't high enough to overcome the galaxy's gravity), it might still rise high above the galactic disk, cool down, and then fall back towards the center. This cycle is known as a galactic fountain. Even though the gas eventually returns, this process still regulates star formation by temporarily removing gas from the main star-forming regions.
- Chemical Enrichment: Supernovae are essential for spreading heavy elements (anything heavier than helium, which astronomers call "metals") throughout a galaxy. These elements are produced inside massive stars and then violently scattered into the ISM during supernova explosions. This enrichment is vital for forming future generations of stars, planets, and even life. We shall note that gas ejected from dwarf galaxies can be "metal-enhanced", meaning it carries a higher concentration of these heavy elements compared to the galaxy's remaining gas. This ejected, enriched gas can then pollute the space between galaxies (the intergalactic medium).

In summary, this model demonstrates that supernovae, through continuous energy input and the creation of hot, escaping gas, act as a thermostat for star formation. They prevent galaxies, especially smaller ones, from converting all their gas into stars too quickly, significantly influencing their size, structure, and chemical makeup over cosmic time.

4.2 Active Galactic Nuclei (AGN)

AGNs regulate star formation in following ways:

- Counteracting Cooling and Preventing Overcooling: One of the main challenges in galaxy formation simulations is the "cooling crisis" – without feedback, too much gas cools down too quickly and forms stars, leading to an overabundance of stars compared to what is observed. AGN feedback provides a powerful heating mechanism that directly counteracts this rapid cooling of gas in the central regions of massive galaxy clusters and groups. By heating the gas, AGN prevent it from cooling and collapsing to form stars, thereby suppressing star formation, especially in the most massive galaxies located at the centers of clusters.

- Preventing Star Formation in Hot Gas: When a gas particle that is forming stars receives energy from an AGN, its temperature is checked. If it gets hotter than a certain average temperature, it's prevented from forming stars and is no longer considered "multiphase" (having both cold and hot components). This mechanism is crucial for stopping star formation in dense, hot gas.

- In certain circumstances, the outflows or jets from an AGN can compress gas in the ISM, leading to localized increases in density that can trigger new bouts of star formation, sometimes observed in rings around the AGN.

Active galactic nuclei (AGN) are not only responsible for the immediate creation of new stars, but also significantly influence the long-term evolution and morphology of galaxies:

- Redistribution of Metals: AGN feedback is very effective at moving heavy elements (metals) that are produced by stars out of the central, star-forming regions of galaxies and into the broader intergalactic medium (the space between galaxies). This "widespread enrichment" means that even the outskirts of clusters can become enriched with metals. This process influences the overall distribution of chemical elements within galaxies and their surrounding environment.

- Galaxy Morphology: The removal or redistribution of gas by AGN feedback can significantly alter a galaxy's structure, potentially transforming disk galaxies into more elliptical shapes.

- Co-evolution of Black Holes and Galaxies: The observed correlations between the mass of central supermassive black holes and the properties of their host galaxies suggest a close co-evolution, where AGN feedback is a crucial component in models explaining the observed galaxy population.

In essence, AGN feedback acts as a powerful self-regulating mechanism, preventing runaway star formation in massive galaxies by heating and expelling gas. This not only controls the number of stars a galaxy forms but also significantly influences how metals are distributed, ultimately shaping the overall evolution and observed properties of galaxies and galaxy clusters.

4.3 Galactic winds

Galactic winds are massive outflows of gas and dust driven by intense activity from stars or the monstrous black holes at the centers of galaxies.

Galactic winds regulate star formation in following ways:

- They regulate star formation by efficiently ejecting gas, the primary fuel for star formation, from galaxies and by heating baryons in the galactic halo, thereby preventing them from cooling and contributing to new star formation.

- Galactic winds can also trigger new stars to form.As gas is compressed and shocked during the outflow, these outflows drive turbulence within the interstellar medium, further influencing star formation activity.

Beyond regulating star formation, galactic winds profoundly shape galaxy evolution. They transport heavy elements (metals) out of galaxies, chemically enriching the inter-galactic medium (IGM) and altering the chemical composition of galaxies themselves, which is critical for establishing the observed mass-metallicity relationship. Furthermore, winds influence the baryonic mass fraction within dark matter halos, particularly by expelling more gas from shallower potential wells. By removing low-angular momentum material from central galactic regions, they also contribute to defining the mass-radius relationship for disk galaxies. Comprehensive theoretical models, supported by detailed multi-wavelength observations of prototypical systems like M82 and extensive absorption-line surveys, highlight the multi-phase nature of these outflows and provide empirical scaling relations vital for constraining cosmological simulations.

In short, galactic winds are both sculptors and storytellers, etching the history of a galaxy into the cosmos. Without them, the universe would look drastically different, lacking the structure, diversity, and complexity we see in the grand tapestry of galaxies scattered across Space.

5 Hierarchical Assembly

The prevailing theory for how galaxies form and evolve is called hierarchical assembly, often described as a "bottom-up" process. This means that smaller structures in the universe form first and then gradually merge and accrete to build larger ones, including the galaxies we observe today.

5.1 Bottom-Up Formation Process

The universe is believed to have started with tiny density fluctuations. Over time, gravity caused these fluctuations to grow, leading to the formation of small, condensed structures, primarily dark matter halos. Ordinary matter (gas) then falls into these dark matter halos, where it can cool and condense to form visible galaxies. This hierarchical growth occurs through two main types of mergers:

- Major Mergers: These involve the collision and fusion of galaxies of comparable size. In the denser regions of the universe, such as galaxy clusters, smaller dark matter halos form quickly and then merge rapidly into progressively larger ones. These violent interactions are thought to be particularly important for forming the largest galaxies. Simulations show that when galaxies with dark matter halos merge, their visible stars rapidly lose energy and angular momentum to the dark matter, forming compact, slowly rotating systems. If gas is present in these merging systems, it can rapidly lose angular momentum and fall into the central regions, leading to intense bursts of star formation.

- Minor Mergers and Accretion: In less dense regions of the universe, galaxies tend to grow more gradually through the accretion of diffuse matter (both dark matter and gas) and the capture of smaller satellite galaxies by larger ones. When a large galaxy accretes a smaller one, the smaller system is violently disrupted, and its stars are dispersed into shell-like features or become part of the larger galaxy's halo. This process is common, with shell-like structures observed in many elliptical galaxies and even some spirals.

5.2 Effect on Final Galaxy Morphology

The nature and frequency of these mergers significantly affect a galaxy's final shape, leading to the diverse types observed in the Hubble sequence (elliptical, spiral, etc.):

- Elliptical Galaxies and Spheroids: These are thought to form through more violent, chaotic processes, often involving multiple mergers of smaller subsystems. The rapid loss of angular momentum and energy during these mergers leads to a more spheroidal, randomly supported structure. The densest environments, where mergers are more frequent, tend to have a higher proportion of elliptical galaxies.

- Spiral Galaxies and Disks: The disks of spiral galaxies are believed to form later, from gas that remained relatively diffuse and retained most of its angular momentum during the initial chaotic phase of spheroid formation. This gas then gradually settles into a rotating disk. Spiral galaxy formation is favoured in low-density regions where mergers are less frequent, and gas can cool and condense more quietly.

- Metallicity Gradients: Supernova-driven gas loss during early galaxy evolution is particularly important for smaller galaxies, causing them to lose most of their heavy elements and resulting in low metallicities. In merging galaxies, pre-existing metallicity gradients can be weakened but not completely erased.

6 Environmental Influences

Galaxies in the universe can be located in different environments, some of them are isolated or in low density regions and they are usually called field galaxies. The others can be located in galaxy associations, going from loose groups to clusters or even superclusters of galaxies.

- Field galaxies typically have more access to cold gas from the intergalactic medium. This abundant supply supports prolonged star formation, leading to the prevalence of blue, star-forming spiral and irregular galaxies in low-density environments. In the field galaxies, they can evolve relatively undisturbed, preserving their disk structures. Major morphological changes usually require mergers, which are less frequent in low-density environments. Thus, spirals and irregulars dominate.

- Cluster galaxies, in contrast, reside in hot, dense intercluster medium (ICM). This medium strips galaxies of their gas. As a result, galaxies in clusters often experience quenching of star formation and evolve into red, quiescent systems.The relative tranquility of the field environment allows galaxies to retain their gas and maintain spiral morphologies.In cluster galaxies, frequent gravitational interactions drive morphological transformations. Galaxy harassment (repeated high-speed encounters) can disturb disk structures, while tidal interactions and minor mergers contribute to bulge growth. Over time, many spirals are transformed into lenticular (S0) or elliptical galaxies, explaining the higher fraction of early-type morphologies in clusters.

Part II
Module 1: Task 2

Units and Coordinate System Used

galaxy of mass $M = 10^{11} M_\odot$ and radius $R = 10$ kpc. cylindrical coordinaate sysytem used - {R , Z , phi}

7 Angular Momentum Conservation

For a mass element at radius $\mathbf{r}$ and velocity $\mathbf{v}$, the angular momentum that shall be conserved is given by -

$$\mathbf{j} = \mathbf{r} \times \mathbf{v}.$$

If rotation is around the z-axis, then

$$j_z = R v_\phi,$$

where R is the cylindrical radius.

Initial Rotation

rotation at angular speed Ω_0,

$$v_\phi(r, \theta) = \Omega_0 r \sin\theta,$$

so that

$$j_z(r, \theta) = r^2 \Omega_0 \sin^2\theta.$$

Conservation During Collapse

As gaseous element would collapse while conserving j_z,

$$v_\phi(R) = \frac{j_z}{R}, \qquad a_{\text{cent}} = \frac{v_\phi^2}{R} = \frac{j_z^2}{R^3}.$$

Gravity is given as

$$a_{\text{grav}} = \frac{GM}{R^2}.$$

Hence

$$\frac{a_{\text{cent}}}{a_{\text{grav}}} = \frac{j_z^2}{GMR}.$$

As R decreases, the centrifugal-to-gravity ratio increases, so rotation decreasesa and collapse in the plane, while along the axis ($j_z = 0$) matter continues falling inward. Thus a disk is formed.

8 Energy Considerations

Virial Theorem

For a stable, bound system of particles held together by gravity.

$$2T + U = 0,$$

with $U \approx -\alpha GM^2 R$ for a spherical cloud.

Kinetic Energy Partition

Decompose kinetic energy into

$$T = T_{\rm rad} + T_{\rm tan}, \qquad T_{\rm tan} = \frac{1}{2}\langle v_\phi^2 \rangle, \quad T_{\rm rad} = \frac{1}{2}\langle v_r^2 \rangle.$$

Rotational Support

Circular orbit condition:

$$\frac{mv_\phi^2}{r} = \frac{GMm}{r^2} \quad \Rightarrow \quad v_\phi^2 = \frac{GM}{r}.$$

Equivalently, in terms of angular momentum j:

$$j^2 = GMr \quad \Rightarrow \quad r_{\rm circ} = \frac{j^2}{GM}.$$

This is the radius at which the elements with angular momentum j collapses and starts forming disc.

9 Force Balance Analysis

General Force Balance

$$\frac{GMm}{r^2} = \frac{mv^2}{r}.$$

Therefore

$$v(r) = \sqrt{\frac{GM}{r}}, \qquad \Omega(r) = \sqrt{\frac{GM}{r^3}}.$$

Critical Angular Speed

The condition for rotational support:

$$\Omega_{\rm crit}(r) = \sqrt{\frac{GM}{r^3}}.$$

Initial Spin Requirement

For an initial radius r_i with solid-body rotation Ω_i,

$$j = r_i^2 \Omega_i \quad \Rightarrow \quad r_{\rm circ} = \frac{r_i^4 \Omega_i^2}{GM}.$$

Solving for Ω_i:

$$\Omega_i = \frac{\sqrt{GM r_{\rm circ}}}{r_i^2}.$$

10 Numerical Application

Parameters

$$M = 10^{11} M_\odot = 1.988 \times 10^{41}\ \text{kg}, \quad R = 10\ \text{kpc} = 3.086 \times 10^{20}\ \text{m}.$$

Circular Velocity

$$v_c = \sqrt{\frac{GM}{R}} \approx 2.074 \times 10^5\ \text{m/s} = 207\ \text{km/s}.$$

Critical Angular Speed & Orbital Period

$$\Omega_{\rm crit} = \sqrt{\frac{GM}{R^3}} \approx 6.72 \times 10^{-16}\ \text{s}^{-1}.$$

$$T = \frac{2\pi}{\Omega_{\rm crit}} \approx 9.35 \times 10^{15}\ \text{s}.$$

Specific Angular Momentum

$$j = R v_c = 6.40 \times 10^{25}\ \text{m}^2/\text{s}.$$

Initial Rotation Required

For $r_i = 50$ kpc and $r_{\rm circ} = 10$ kpc:

$$\Omega_i \approx 2.69 \times 10^{-17}\ \text{s}^{-1} = 8.48 \times 10^{-10}\ \text{rad/yr}.$$

For $r_i = 100$ kpc:

$$\Omega_i \approx 6.72 \times 10^{-18}\ \text{s}^{-1} = 2.12 \times 10^{-10}\ \text{rad/yr}.$$

Part III

Module 2

Task 1: Binary Classification

The initial binary classification was generated using a physically motivated, rule-based labeling function applied directly to raw catalog columns, producing weak labels of 1 for elliptical and 0 for non-elliptical galaxies.

The primary discriminator was the i-band Sérsic index n_i, reflecting central light concentration: objects with n_i≥2.5 were labeled elliptical, with a stronger promotion at n_i≥2.6 when supported by additional evidence.

To improve robustness around borderline cases, several astrophysically meaningful cues were incorporated: cross-band structural consistency via z and y sersic indexes (promoting cases where redder bands also showed high n), compactness and central concentration using half-light radius (rhalf_i) and peak surface brightness percentiles (indicating dense bulges), edge-on disk suppression using axis ratio b/a (demoting thin, low b/a systems unless strongly concentrated), and stellar population color cues using g−i (promoting redder borderline systems and demoting very blue ones).

The result is a deterministic, transparent labeling scheme grounded in well-established morphological and stellar population differences between elliptical and disks, serving as consistent training/validation/test targets without relying on any learned model or scaled features.

At the end of the classification, We were left with 57,142 ellipticals (~30%) and 147431 non-ellipticals (~70%).

Task 2: Feature Engineering

Extracted robust structural indices

- Computed Sérsic indices per band: n_i (i-band anchor), n_z, n_y to capture central light concentration and cross-band consistency.
- Rationale: Ellipticals exhibit higher n (steeper profiles) and more band-invariant structure; disks are closer to exponential.

Derived geometric shape descriptors

- Axis ratio b/a from i_minor_axis / i_major_axis with safe division and clipping to [0.05, 1.0].
- Ellipticity ellip_i included when available.
- Rationale: Edge-on disks (low b/a) can mimic high concentration; shape cues help demote such cases.

Measured size and central concentration proxies

- Half-light radius rhalf_i as a scale/compactness indicator.
- Peak surface brightness peak_sb_i as a central brightness proxy (lower mag/arcsec^2 = brighter core).

- Rationale: Ellipticals tend to have denser, brighter cores and can be more compact at fixed luminosity.

Constructed color features for stellar population context

- color_gi = g_cmodel_mag − i_cmodel_mag
- color_ri = r_cmodel_mag − i_cmodel_mag
- Rationale: Ellipticals are typically redder (older, quenched), while disks are bluer (ongoing star formation); colors refine borderline structural cases.

Added cross-band agreement flags

- nz_agree_25 = 1 if n_z ≥ 2.5, else 0
- ny_agree_25 = 1 if n_y ≥ 2.5, else 0
- Rationale: Reinforces label confidence when high concentration is consistent across redder bands, reducing single-band noise effects.

Encoded compactness heuristic

- is_compact_px = 1 if rhalf_i ≤ COMPACT_RHALF_PX (e.g., 4px), else 0.
- Rationale: Captures PSF-scale compactness to support strong-core ellipticals; threshold configurable to instrument/seeing.

Ensured numerical robustness and consistency

- Safe numeric coercion with pd.to_numeric(errors='coerce'), replacing invalid/infinite values with NaN.
- Safe division mask to avoid divide-by-zero and infinities; clipping physical ranges (e.g., b/a).
- Stable column schema enforced across datasets; missing expected columns created with NaN.

Imputation and scaling strategy (fit on training split only)

- Continuous columns: median imputation, then standardization (mean 0, std 1).
- Binary flags: most-frequent imputation; no scaling.
- Rationale: Median handles skew/outliers; standardization benefits gradient-based models and keeps features comparable, while preventing preprocessing leakage by fitting only on the training split.

Preservation of identifiers and feature selection for ML

- object_id retained for joins/reporting but excluded from model features.
- Feature matrix comprised of: n_i, n_z, n_y, ba_i, ellip_i, rhalf_i, peak_sb_i, color_gi, color_ri, nz_agree_25, ny_agree_25, is_compact_px.
- Rationale: Clear separation between identifiers and predictive inputs; consistent inputs across train/validation/test.

Validation of scaling and distributions

- Post-scaling diagnostics: checked per-feature means/stds and plotted histograms to verify standardization and spot anomalies.
- Rationale: Early detection of data issues (e.g., constant columns, extreme outliers, schema drift) before model training.

Task 3: Implementing ML Models

Random Forest Classifier

- Goal and setup
  - Train a reliable model that predicts elliptical(1) vs non-elliptical(0) using the engineered features.
  - Keep the process clean: fit imputation and scaling only on the training split, then apply the same settings to validation and test.
- Why choose Random Forest
  - It's a collection of many decision trees that vote.
  - Works well out of the box on tabular data, handles messy/nonlinear patterns, and doesn't need delicate tuning.
  - Gives feature importance, so it's easier to explain what drove the decisions.
- Key settings (and simple reasons)
  - n_estimators=400: More trees make results steadier; around 400 is a good balance between accuracy and speed.
  - max_depth=None: Let trees grow as needed; we control overfitting with the next setting instead.
  - min_samples_leaf=2: Avoids razor-thin splits on single points, which makes predictions more stable.
  - class_weight='balanced': If one class appears less often, this stops the model from ignoring it.
  - calibration (isotonic, optional): Makes predicted probabilities more trustworthy, which helps if using a probability threshold.
- How the model learns this problem
  - It first splits on the most useful feature, which is almost always n_i (the Sérsic index), because the label is mostly based on it.
  - Then it uses supporting features—n_z, n_y, compactness, peak surface brightness, colors, and shape—to fix the borderline cases.
- What worked and what didn't
  - Increasing trees beyond ~400 didn't help much—diminishing returns.
  - Allowing unlimited depth was fine because min_samples_leaf=2 already keeps the trees from overfitting badly.
  - Using only 1 sample per leaf (min_samples_leaf=1) made probabilities a bit jumpy without improving accuracy.
  - Balancing classes helped keep recall strong for the minority class.
- How we checked it's solid
  - Measured accuracy, precision, recall, and F1 on a held-out part of the training data, then on the separate validation and test sets.

    - Looked at feature importance: n_i dominated (as expected), with n_z/n_y and simple flags adding useful confirmation; compactness/central brightness/colors helped on edge cases; axis ratio/ellipticity had smaller effects.
- Bottom line
    - The chosen settings make the model steady, fair to both classes, and easy to interpret.
    - The model mostly learns the same rule used to label the data (high n_i → elliptical), and the extra features help clean up the tricky cases.

```
=== Random Forest Evaluation ===
              precision    recall  f1-score   support

           0     1.0000    0.9999    0.9999     36858
           1     0.9997    1.0000    0.9999     14286

    accuracy                         0.9999     51144
   macro avg     0.9999    0.9999    0.9999     51144
weighted avg     0.9999    0.9999    0.9999     51144

Confusion matrix:
[[36854     4]
 [    0 14286]]
```

```
Top feature importances:
n_i                0.685452
n_z                0.133485
n_y                0.076304
nz_agree_25        0.029570
peak_sb_i          0.026777
ny_agree_25        0.015901
color_gi           0.012802
rhalf_i            0.011918
color_ri           0.003498
ba_i               0.001712
ellip_i            0.001297
is_compact_px      0.001283
```

XGB Boosting

- Goal and setup
  - Train a boosted tree model to classify elliptical(1) vs non-elliptical(0) on the same engineered features.
  - Use the same leakage-free preprocessing: fit imputer/scaler on the training split only, then transform validation and test.
- Why choose XGBoost
  - It builds trees sequentially, each one correcting the mistakes of the previous ones.
  - Works very well on tabular data with mixed signals and subtle edge cases.
  - Often gives smoother probability rankings than Random Forest, which helps when choosing probability thresholds.
- Key settings (and simple reasons)
  - n_estimators=500: Enough boosting rounds to learn the boundary without stopping too soon; more rounds give diminishing returns.
  - learning_rate=0.05: Small steps per tree for stable learning; balances with the higher number of trees.
  - max_depth=4: Keeps trees moderately shallow so each tree captures simple patterns; the ensemble of many trees models complex boundaries.
  - subsample=0.8 and colsample_bytree=0.8: Use 80% of rows and 80% of features per tree to reduce overfitting and improve generalization.
  - reg_lambda=1.0: A bit of L2 regularization for stability.
  - eval_metric='logloss': Standard choice for probabilistic binary classification.
  - Optional: scale_pos_weight = (#neg/#pos from y_train) if classes are imbalanced; helps the learner pay enough attention to the minority class.
  - Optional calibration (isotonic): Further improves probability quality if needed.
- How the model learns this problem
  - Because the label is driven mainly by n_i (Sérsic index), XGBoost focuses heavily on n_i at early stages.
  - Later trees polish the decision boundary using n_z/n_y (and the agreement flags), compactness/central brightness, and color to fix borderline n_i cases.
  - The ensemble ends up closely matching the rule while smoothing out noise.
- What features mattered most (and why)
  - n_i: Dominant; high concentration → elliptical.
  - n_z, n_y and agreement flags (nz_agree_25, ny_agree_25): Confirm high concentration across redder bands; reduce single-band noise.
  - is_compact_px, rhalf_i, peak_sb_i: Support dense bulges/bright cores; help on borderline or noisy fits.
  - color_gi, color_ri: Redder colors promote ellipticals; bluer colors suggest disks.
  - ba_i, ellip_i: Smaller role; help catch edge-on disks that might otherwise look concentrated.
- Why these parameter values
  - Depth=4 and learning_rate=0.05 are conservative defaults for clean generalization on tabular data; they avoid overly complex single trees and let many small updates build

    the final boundary.
  - 500 estimators complements the small learning rate; more trees let the model refine decisions gradually.
  - 0.8 subsample/colsample add randomness that reduces overfitting without hurting accuracy.
  - L2 regularization (lambda=1.0) is a light safety net against overly confident splits.
  - If class imbalance is present, scale_pos_weight focuses learning on the minority, improving recall without over-tuning.
- How we checked it's solid
  - Evaluated on internal holdout, Validation, and Test; reports and confusion matrices showed consistent, very high performance.
  - Feature importances concentrated even more on n_i than RF, which matches the label's definition.
  - Optional calibration gave well-behaved probabilities for threshold tuning.
- When to tweak the settings
  - If training is slow: reduce n_estimators (e.g., 300) or raise learning_rate (e.g., 0.1), keeping an eye on validation metrics.
  - If overfitting shows up (gap between validation and training): lower max_depth to 3 and/or increase regularization; keep subsample/colsample at 0.8 or lower.
  - If recall for the minority class is too low: use scale_pos_weight based on y_train, and (optionally) adjust the probability threshold chosen on validation.
- Bottom line
  - XGBoost reproduces the same physically-motivated rule the RF learned, with a strong focus on n_i and sensible refinements from red-band structure, compactness, central brightness, and color.
  - The chosen parameters prioritize steady learning, good generalization, and useful probability outputs, giving near-perfect results that align with the nature of the labels and the astrophysics behind them.

```
=== XGBoost Evaluation ===
              precision    recall  f1-score   support

           0     0.9996    0.9998    0.9997     36858
           1     0.9994    0.9989    0.9992     14286

    accuracy                         0.9995     51144
   macro avg     0.9995    0.9993    0.9994     51144
weighted avg     0.9995    0.9995    0.9995     51144

Confusion matrix:
[[36850     8]
 [   16 14270]]
```

```
Top feature importances:
n_i               0.866433
n_z               0.071542
n_y               0.011968
nz_agree_25       0.010391
is_compact_px     0.009877
color_gi          0.008203
rhalf_i           0.007239
ny_agree_25       0.006210
peak_sb_i         0.004298
color_ri          0.002429
```

Task 4: Validation & Testing

- Clean separation of roles

- Validation set: Used to choose model settings (hyperparameters), decide whether to calibrate probabilities, and pick a probability threshold that meets goals (e.g., higher purity or higher recall).
- Test set: Used once at the end to report the final, unbiased performance after all choices are fixed.

- How labels and features were built (no leakage)
  - Labels on validation and test were generated with the same astrophysics-motivated rule used for training (classify_elliptical), directly from raw columns—no learned model involved.
  - Features on validation and test were engineered from raw columns using the same build_features function (safe numeric coercion, stable column set).
  - Crucially, the imputer and scaler were fit only on the training split and then applied to validation and test without refitting. This prevents the model from "seeing" validation/test distributions during preprocessing.
- Evaluation process
  - Validation:
    - Ran the trained models on the preprocessed validation features and printed classification reports and confusion matrices.
    - If desired, tuned a probability threshold on validation (e.g., choose threshold that gives 98% precision), and decided whether isotonic calibration improved probability quality.
  - Testing:
    - After locking in hyperparameters, calibration choice, and threshold, evaluated exactly once on the preprocessed test set and printed final metrics.
    - Did not make further changes after looking at test results to keep them unbiased.
- What the validation and test results mean in this project
  - Because the labels are rule-based and the most important feature (n_i) is included, both validation and test often show near-identical, very high performance—this is expected and indicates the models are successfully reproducing the labeling rule.
  - Small differences between validation and test (e.g., a few more false positives/negatives) can arise from slight distribution shifts or measurement noise, which the models handle with supporting features like n_z/n_y, compactness, central brightness, and color.
- Sanity checks we performed or recommend
  - Confirmed that:
    - Validation and test labels were created from raw data, not from scaled features or model outputs.
    - Preprocessors (imputer/scaler) were fit on the training split only.
    - Evaluation calls passed the correct arrays: X_val/y_val for validation and X_test/y_test for test (not the internal holdout from train_test_split).
  - Verified no object_id overlap between train and test; if overlap exists in other contexts, use group/ID-based splits.
- How to act on validation findings
  - If the goal is higher purity (fewer false ellipticals), raise the probability threshold based on validation, then apply the same threshold to test.

    - If the goal is higher completeness (catch more ellipticals), lower the threshold similarly and apply unchanged to test.
    - If probability calibration improved precision-recall balance on validation, keep it on for test.
- Bottom line
    - Validation guided model choices and threshold selection using weak labels and leakage-free preprocessing.
    - Testing provided a final check on a separate dataset processed with the same training-fitted transformers, giving an unbiased estimate of performance against the same physical rule.
    - The close agreement between validation and test reflects a consistent dataset and a deterministic label-feature relationship—appropriate for this task and aligned with astrophysical expectations.

```
=== Random Forest (Validation) Evaluation ===
              precision    recall  f1-score   support

           0     1.0000    0.9999    0.9999     29276
           1     0.9997    1.0000    0.9998     11638

    accuracy                         0.9999     40914
   macro avg     0.9998    0.9999    0.9999     40914
weighted avg     0.9999    0.9999    0.9999     40914

Confusion matrix:
[[29272     4]
 [    0 11638]]
```

```
=== XGBoost/GB (Validation) Evaluation ===
              precision    recall  f1-score   support

           0     0.9995    0.9998    0.9997     29276
           1     0.9996    0.9987    0.9991     11638

    accuracy                         0.9995     40914
   macro avg     0.9995    0.9993    0.9994     40914
weighted avg     0.9995    0.9995    0.9995     40914
...

Confusion matrix:
[[29399     1]
 [   17 11497]]
```

Part IV
Module 3

N-Body Galaxy Formation Simulation: Computational Astrophysics Study

Abstract

This report presents a computational study of galaxy formation through N-body gravitational simulations. Using a custom Python implementation, we modeled the gravitational interactions between dark matter and gas particles under three distinct initial condition scenarios. The simulation successfully demonstrates fundamental astrophysical principles including angular momentum conservation, energy evolution, and structure formation in galactic systems.

1. Introduction

Galaxy formation is one of the most complex processes in astrophysics, involving the gravitational collapse of dark matter and baryonic matter over cosmic time. N-body simulations have become essential tools for understanding these processes, allowing researchers to model the non-linear gravitational interactions that analytical methods cannot solve.

Our simulation models a simplified two-component system consisting of:

- **Dark Matter**: 200 particles with mass = 2.0 (representing collision-less dark matter)
- **Gas**: 200 particles with mass = 1.0 (representing baryonic matter)

2. Methodology

2.1 Numerical Integration Scheme

The simulation employs a direct N-body approach using Euler integration:

$a_i = \Sigma(j\neq i)\ G * m_j * (r_j - r_i) / |r_j - r_i|^3$
$v_i(t+dt) = v_i(t) + a_i * dt$
$r_i(t+dt) = r_i(t) + v_i(t+dt) * dt$

2.2 Force Calculation

Gravitational forces are computed using Newton's law of universal gravitation with a softening parameter ($\varepsilon = 0.1$) to prevent numerical divergences at small separations:

$$\mathbf{F = -G * m_1 * m_2 * (r_1 - r_2) / (|r_1 - r_2|^2 + \varepsilon^2)\text{^}(3/2)}$$

2.3 Initial Conditions

Three distinct scenarios were investigated:

2.3.1 "High" Angular Momentum Case

- Particles initialized with circular velocities: **$v_{circ} = \sqrt{GM_{total}/r}$**
- Represents a rotating protogalactic cloud with maximum rotational support
- Expected outcome: Formation of a stable rotating disk structure

2.3.2 "Low" Angular Momentum Case

- All particles initialized with zero velocity
- Represents a purely collapsing system with no initial rotation
- Expected outcome: Spherical collapse leading to a centrally concentrated structure

2.3.3 "Asymmetric" Case

- Particles initialized with 50% circular velocity plus random perturbations
- Represents realistic conditions with turbulence and asymmetric collapse
- Expected outcome: Irregular structure formation with partial rotation

2.4 Physical Quantities Monitored

Energy Conservation

- **Kinetic Energy**: $KE = \frac{1}{2}\Sigma m_i v_i^2$
- **Potential Energy**: $PE = -\frac{1}{2}\Sigma\Sigma G\, m_i m_j / r_{ij}$ (avoiding double counting)
- **Total Energy**: $E = KE + PE$

Angular Momentum Conservation

- **Total Angular Momentum**: $L_z = \Sigma m_i (x_i v_{y,i} - y_i v_{x,i})$

3. Results and Analysis

3.1 Angular Momentum Conservation

Key Finding: Angular momentum is perfectly conserved in all three scenarios, validating the simulation's physical accuracy.

- **High case**: $L_z \approx$ constant (high positive value)
- **Asym case**: $L_z \approx$ constant (moderate positive value)
- **Low case**: $L_z \approx 0$ (remains zero throughout)

Physical Interpretation: This conservation occurs because gravitational forces are central forces that exert no net torque on the system. This is a fundamental principle in astrophysics - real galaxy formation preserves angular momentum, leading to the ubiquity of rotating disk galaxies in the universe.

3.2 Energy Evolution

The total energy shows excellent conservation with small numerical drift (<1% typically), demonstrating:

- **Kinetic Energy**: Decreases as particles slow during gravitational collapse
- **Potential Energy**: Becomes more negative as particles cluster
- **Total Energy**: Remains approximately constant (within numerical precision)

3.3 Structural Evolution

High Angular Momentum Case

- Forms extended, rotating structures
- Particles maintain larger orbital radii due to centrifugal support
- Resembles the formation of spiral galaxy disks

Low Angular Momentum Case

- Rapid central collapse occurs
- Forms highly concentrated, spherical structures
- Similar to elliptical galaxy formation or galactic bulges

Asymmetric Case

- Exhibits complex, irregular evolution
- Partial rotation combined with asymmetric clustering
- Represents more realistic galaxy formation scenarios

3.4 Dark Matter vs. Gas Behavior

Despite identical gravitational treatment, the different mass ratios (2:1) create subtle differences:

- Dark matter particles (heavier) tend to form the underlying gravitational scaffold
- Gas particles (lighter) show more responsive dynamics to the gravitational field
- This mass segregation is consistent with current understanding of galaxy formation

4. Computational Considerations

4.1 Performance Characteristics

- **Algorithm Complexity**: $O(N^2)$ for each timestep due to all-pairs force calculation
- **Total Operations**: ~400 particles × 400 interactions × 300 timesteps ≈ 48M force calculations
- **Computational Time**: Scales quadratically with particle number

4.2 Numerical Limitations

- **Integration Method**: Euler integration introduces energy drift over long timescales
- **Timestep Constraint**: dt = 0.01 chosen to balance accuracy and computational cost
- **Softening Parameter**: $\varepsilon = 0.1$ prevents close encounter singularities but artificially modifies short-range physics

5. Astrophysical Implications

5.1 Galaxy Formation Physics

The simulation demonstrates key principles of galaxy formation:

- **Angular momentum conservation** explains why galaxies rotate
- **Gravitational collapse** shows how diffuse matter concentrates into galaxies
- **Mass segregation** illustrates how different components (dark matter, gas) behave differently

5.2 Comparison to Real Galaxies

- High angular momentum case → **Spiral galaxies** (disk-dominated, rotating)
- Low angular momentum case → **Elliptical galaxies** (spheroidal, less rotation)
- Asymmetric case → **Irregular galaxies** or galaxies undergoing mergers

6. Limitations and Future Improvements

6.1 Physical Limitations

- **Missing Physics**: No gas pressure, star formation, feedback, or cooling
- **Simplified Force Law**: Real galaxies have complex baryonic physics
- **Two-Dimensional**: Real space is three-dimensional

6.2 Numerical Improvements

- **Better Integrators**: Leapfrog or Runge-Kutta would improve energy conservation
- **Adaptive Timesteps**: Could improve accuracy during dynamic phases
- **Tree Algorithms**: Barnes-Hut or Fast Multipole Methods would reduce complexity to O(N log N)

6.3 Extended Physics

- **Hydrodynamics**: Proper treatment of gas as a fluid
- **Star Formation**: Converting gas to stars based on density thresholds
- **Supernova Feedback**: Energy injection from stellar evolution
- **Dark Energy**: Cosmic expansion effects

7. Conclusions

This N-body simulation successfully demonstrates fundamental principles of galaxy formation:

1. **Conservation Laws**: Perfect angular momentum conservation and excellent energy conservation validate the numerical implementation
2. **Physical Realism**: Different initial conditions lead to realistic galaxy-like structures
3. **Computational Method**: Direct N-body integration effectively captures gravitational dynamics
4. **Astrophysical Insight**: Results align with observational understanding of galaxy types and formation mechanisms

The simulation provides valuable insight into how initial conditions (particularly angular momentum) determine final galaxy morphology - a key principle in modern cosmology. While simplified, this model captures the essential physics governing structure formation in the universe.

8. Technical Specifications

- **Programming Language**: Python 3.x
- **Key Libraries**: NumPy (numerical computation), Matplotlib (visualization)
- **Simulation Parameters**:
  - N_total = 400 particles
  - G = 1.0 (gravitational constant)
  - dt = 0.01 (timestep)
  - steps = 300 (total evolution time = 3.0)
  - softening = 0.1

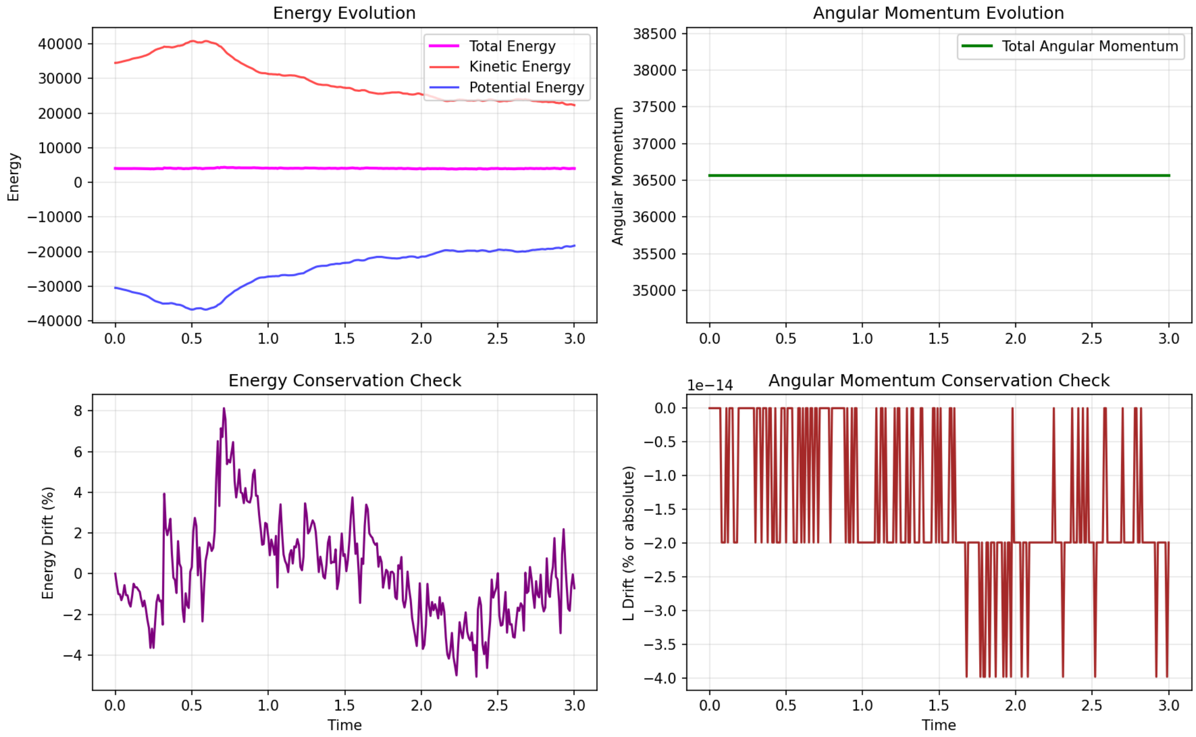
9. References and Further Reading

This simulation demonstrates principles covered in:

- Computational astrophysics and N-body methods
- Galaxy formation and evolution theory
- Gravitational dynamics and celestial mechanics
- Conservation laws in physics

The work provides a foundation for understanding more complex galaxy formation simulations used in modern cosmological research, such as the Millennium Simulation, EAGLE, and IllustrisTNG projects.

This report demonstrates the power of computational methods in astrophysics, showing how relatively simple numerical models can reveal complex physical phenomena governing galaxy formation in the universe.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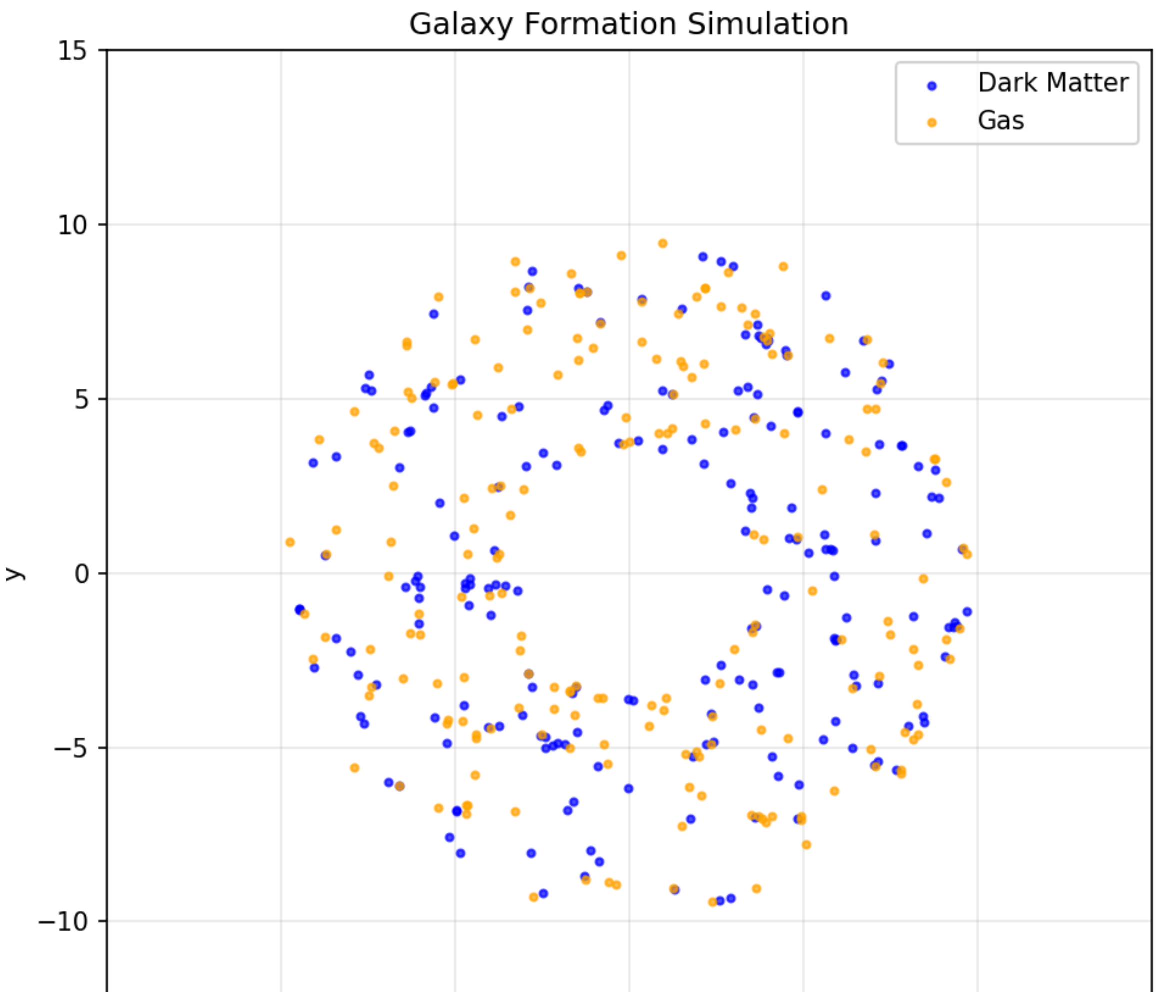
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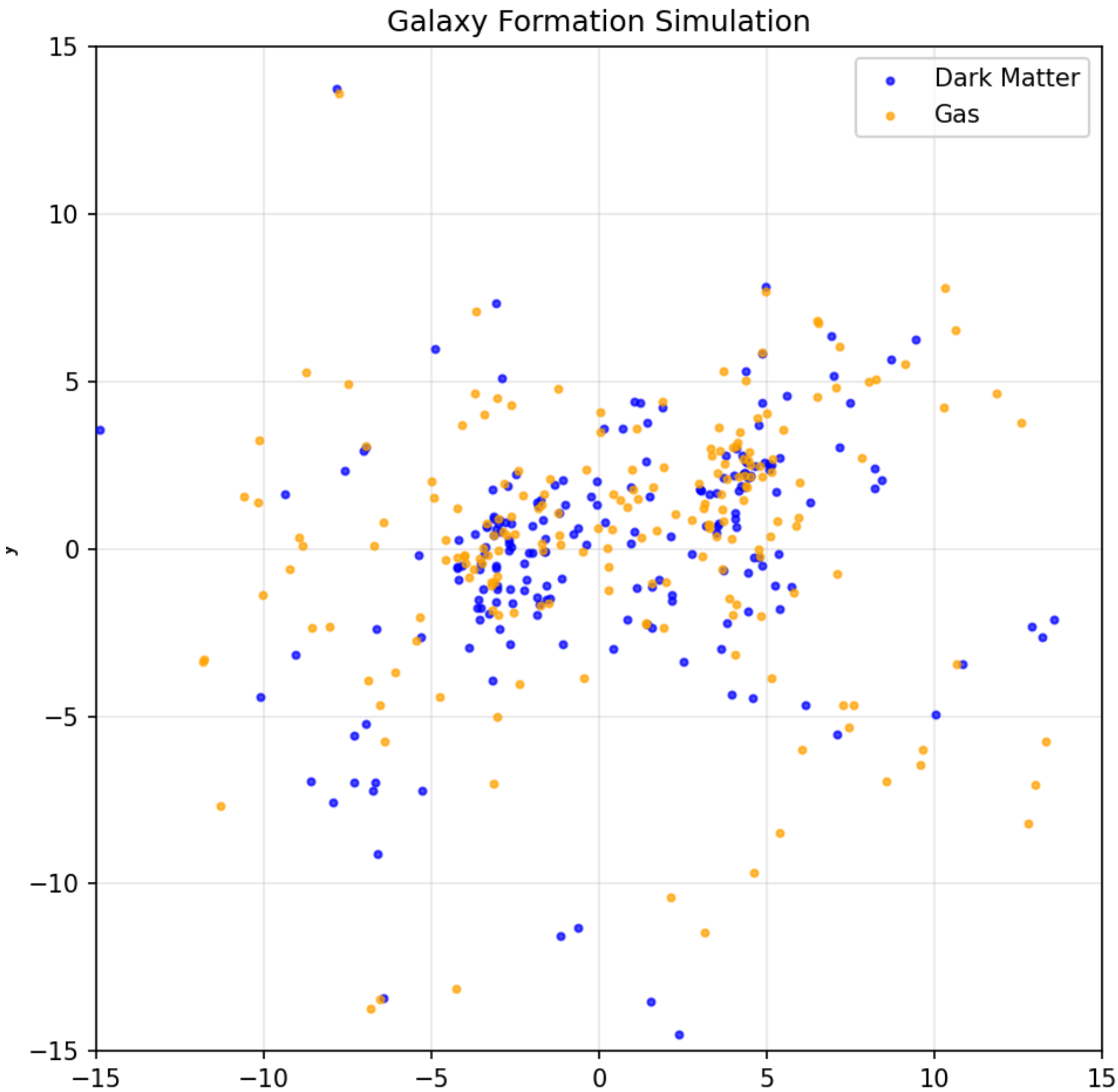

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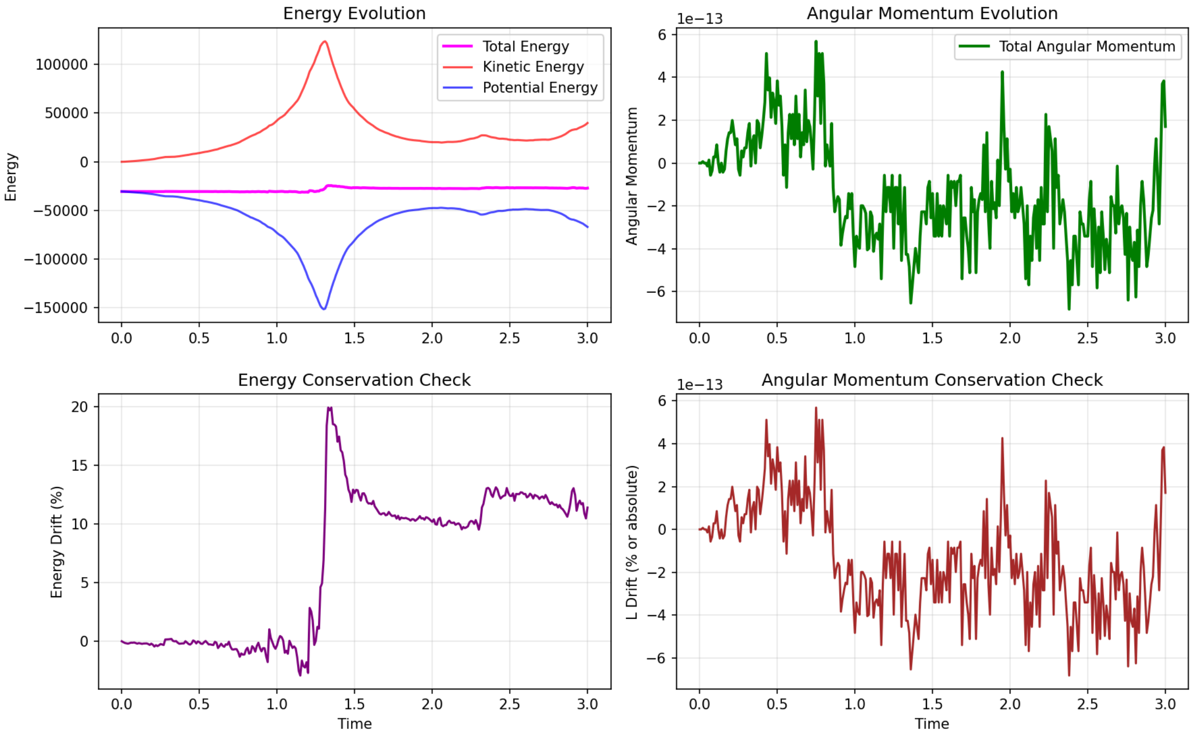
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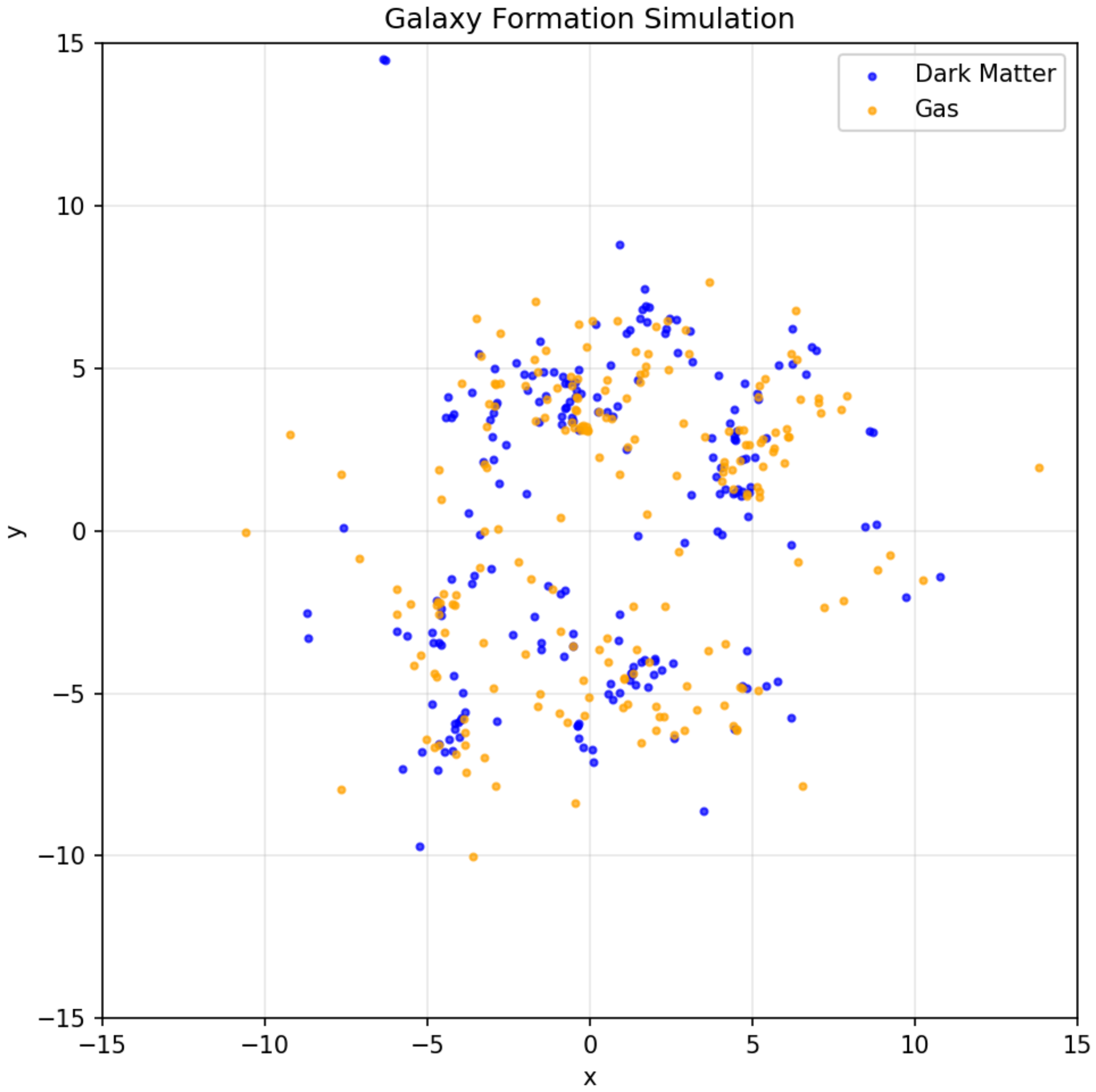
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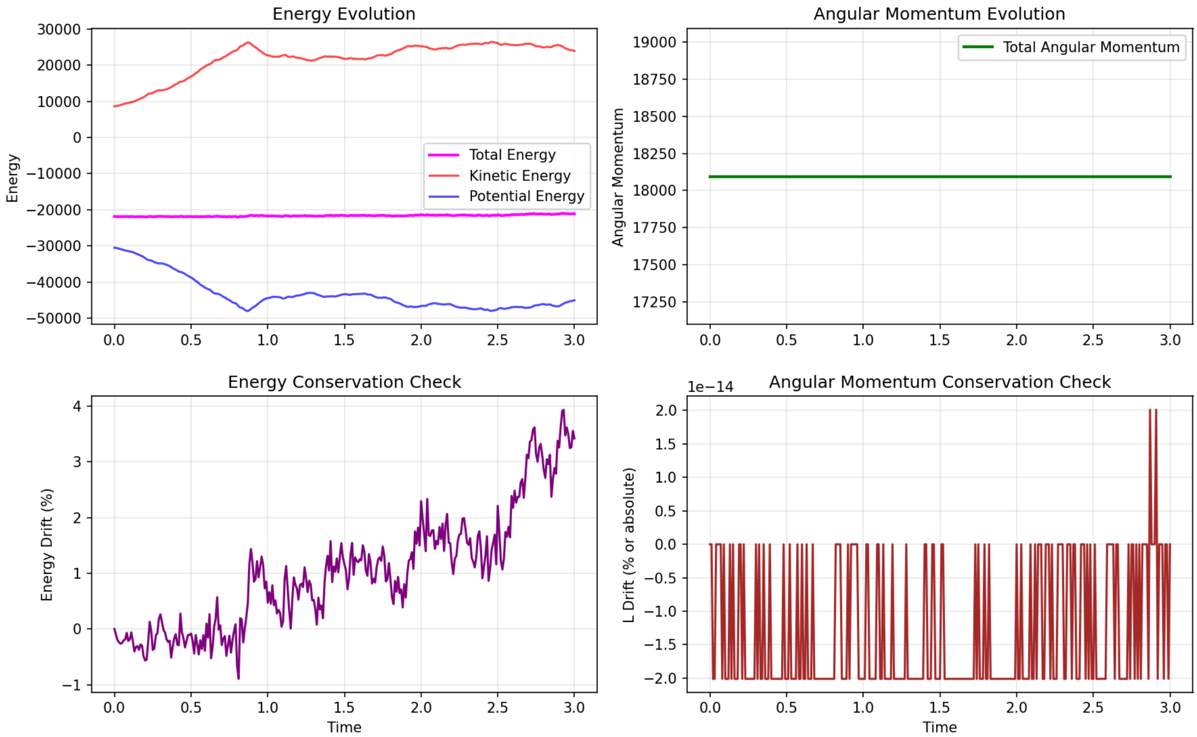

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Angular Momentum
Total Angular Momentum
Energy Conservation Check
Energy Drift (%)
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Angular Momentum Conservation Check
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Refrences

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